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AVDASI-3: Sustainable Aviation & Ethics

Lecture-9: Costing / commercial aspects of aircraft design

Samudra Dasgupta

Senior Lecturer – CADE, University of Bristol

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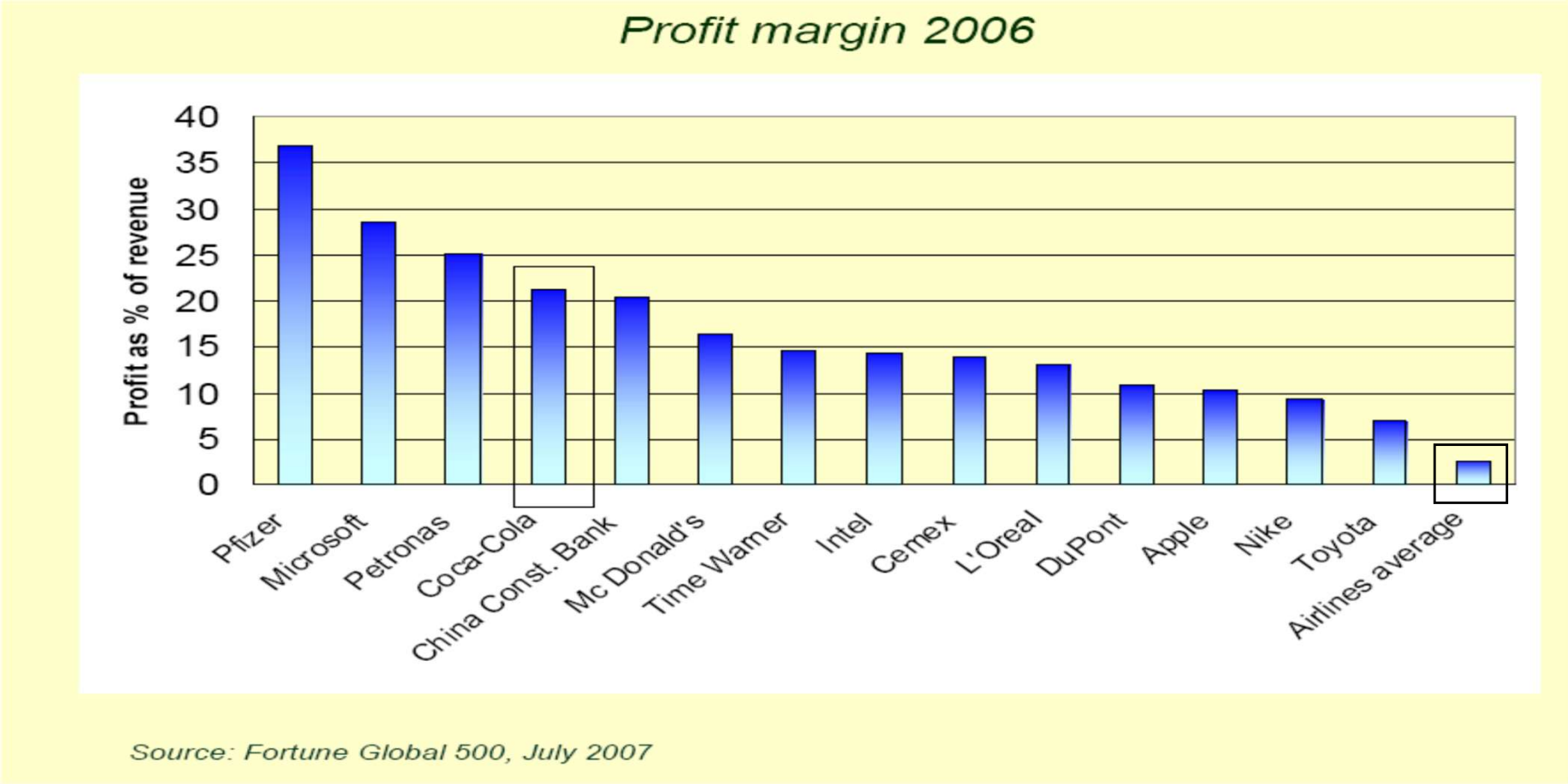
Typical industrial profit margins



Operating Costs - Different Methods

- **Initial Cost**
  - Only considers the initial acquisition costs
    - ☺ Very simple
    - ☹ Doesn't account for day-to-day running costs or reliability
- **Cash, Direct and Indirect Operating Costs (COC, DOC & IOC)**
  - Widely used by civil aerospace industry to evaluate total costs per trip
    - ☺ Accounts for all costs
    - ☺ Suits civil market conditions
    - ☹ Can be biased according to needs
- **Life Cycle Cost (LCC)**
  - Typically used by military to evaluate total program lifetime cost
    - ☺ Considers everything over entire program lifetime
    - ☹ Complex models
    - ☹ Military method details can be classified

Operating Costs - COC, DOC & IOC (1/2)

Direct Operating Cost (DOC)

- Financial Costs
  - Depreciation
  - Interest
  - Insurance

Cash Operating Cost (COC)

- Flying Costs
  - Fuel
  - Landing fees
  - Cockpit crew
  - Cabin crew
  - Navigation charges
- Maintenance Costs
  - Airframe
  - Engines

Dependent on aircraft design

Total Operating Cost (TOC)

Indirect Operating Cost (IOC)

- Ground Property & Equipment
  - Depreciation & Maintenance
- Administration & Sales
  - Servicing administration
  - Reservations & sales
  - Advertising & publicity
  - General
- Servicing
  - Passenger services
  - Aircraft services
  - Traffic services

Dependent on airline operations

Operating Costs - COC, DOC & IOC (2/2)

- **Cash Operating Cost (COC):**
  - Flight-related costs
    - ☺ Highlights aircraft use and variable cost trends – Useful to airlines
    - ☹ Doesn't account for aircraft cost - If used as the target function, it drives design to a high-tech (high cost) solution to reduce fuelburn
- **Direct Operating Cost (DOC):**
  - COC + Finance-related costs (aircraft price or cost)
    - ☹ Large price/cost component masks flight-related cost trends which are important for airlines
    - ☺ Realistically accounts for the cost of aircraft design and technology
- **Indirect Operating Costs (IOC):**
  - Airline infrastructure costs
    - ☹ Highly airline dependent – No reliable quantitative method

Operating Costs - Different DOC Methods

There are many methods of calculating DOC with different financial assumptions and component breakdowns:

Manufacturers:
(Confidential)

- AI (C) — Airbus (Commercial)
- AI (P) — Airbus (Project)
- OPCOST — Boeing
- RR — Rolls Royce

Airlines:
(Confidential)

- Lufthansa
- British Airways, etc.

Trade groups:
(Public domain)

- ATA — Air Transport Association (1967)
- AEA — Association of European Airlines (recommended – refer async document)
- CEE — Committee Etude Economique
- FEAT — Future Economic Advanced Transport WG

Self-proclaimed experts:

- "www.fred-in-a-shed.com"
- "www.university-of-somewhere-you've-never-heard-of.ac"

Always use the same method when comparing different aircraft!

AEA Method - Inputs, Assumptions & Results

Inputs

- Mission data:
  - Stage Length (NM)
  - Block Fuel [BF] (lb)
  - Block Time [BT] (hr)
  - Passengers [Pax]
- Weight data:
  - MTOW
  - MWE
  - Engine Weight
- Engine parameters:
  - Number of Engines [NE]
  - SLST [T]
  - Bypass Ratio [BPR]
  - Overall Pressure Ratio [OPR]
  - No. of compressor stages [NC]
- Price data:
  - Engine Price [ENP] ($)
  - Manufacturers Study Price [MSP] ($)
  - Airframe Cost [AFC] ($)
  - Fuel Price ($/USgal)

Assumptions

- Fuel Density = 6.7 lb/USgal
- Labour Rate [R] = 66 $/hr

AEA DOC Method

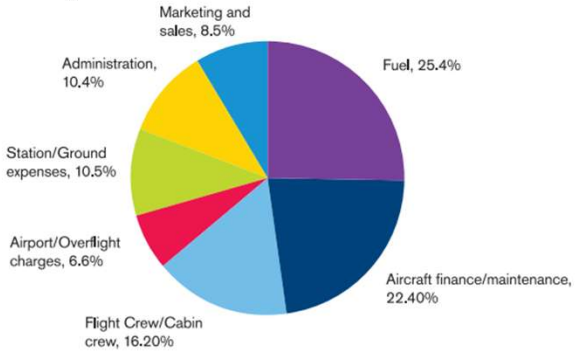
Results

- Financial Costs:
  - Depreciation [DEP]
  - Interest [INT]
  - Insurance [INS]
- Maintenance Costs:
  - Airframe Maintenance [AMC]
  - Engine Maintenance [EMC]
- Flight Costs:
  - Cockpit Crew [CPC]
  - Cabin Crew [CAC]
  - Navigation Charges [NAV]
  - Landing Fees [LAF]
  - Fuel [FUE]

(All costs calculated as $/trip)

DOC

COC



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Please refer asynchronous document

AEA Method for DOC calculations

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Low-cost vs traditional carriers

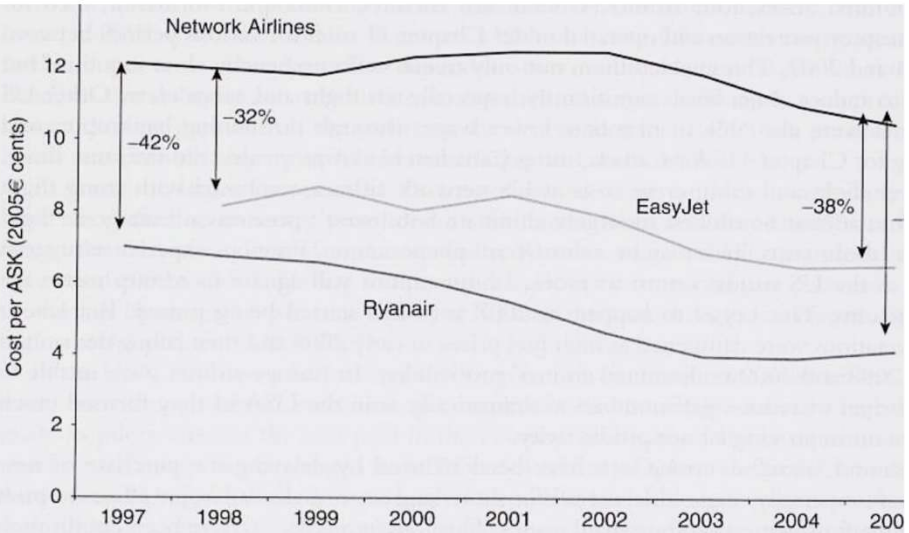

Figure 6.3 Comparing unit costs of European network and low-cost airlines, 1997–2005.
Source: IATA (2007).

Ref: Doganis 2009

ITEM	US Airways	Southwest	Difference (Cents)	%
Labour and Benefits	5.08	2.81	2.27	80.8
Lease and Depreciation	1.65	1.02	0.63	61.8
Maintenance, Materials and outside repairs	0.71	0.65	0.06	9.2
Landing Fees and Outside services	1.19	0.67	.052	77.6
Marketing, Commissions and food	1.20	0.63	0.57	90.5
Fuel	1.69	1.38	0.31	22.5
Other	0.89	0.53	0.36	67.9
Total	**12.41**	**7.69**	**4.72**	**61.4**

–Morten, Beyer and Agnew (The International Aviation Oracle 8 8, 9 Sept 2001, p3

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Thank you

samudra.dasgupta@bristol.ac.uk

https://in.linkedin.com/in/samudradg

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